

to *yahoo.com*. It is hard to remember the IP address, and the IP can change anytime. Also, there will be more than one IP address for a site for load balancing reasons. The example given below returns six IP addresses for Yahoo and the browser needs to choose one.

```
% dig yahoo.com
;; ANSWER SECTION:
yahoo.com. 600 IN A 74.6.231.20
yahoo.com. 600 IN A 74.6.143.26
yahoo.com. 600 IN A 74.6.231.21
yahoo.com. 600 IN A 98.137.11.163
yahoo.com. 600 IN A 74.6.143.25
yahoo.com. 600 IN A 98.137.11.164
```

This is why we have domain names. The DNS or domain name system is a hierarchical and decentralised naming system that works just like a phone book. IP protocol uses DNS to get the destination IP address. An excellent explanation of this is given at <https://www.open.edu/openlearncreate/mod/oucontent/view.php?id=129584&printable=1>

IPSec

Internet protocol security (IPSec) is a network protocol suite that authenticates and encrypts each packet of data to provide secure encrypted communication between two computers over an IP network.

IPSec is a connection-oriented suite of protocols that is used to establish mutual authentication between computers at the beginning of a communications session and to negotiate cryptographic keys during the session. It uses the keys to encrypt the data for communication.

This protocol is commonly used in virtual private networks (VPNs), helping data transfer over a public network but with encryption of the IP packets and also with authentication of the source of the packets. IPSec usually uses port 500.

IPSec ensures the following.

- **Confidentiality:** Protects data from

being accessed by unauthorised persons. Only the sender and receiver can read the data, as it uses encryption.

- **Integrity:** Ensures data is not being tampered intentionally or unintentionally. The hash value of the data from the sender is matched with that of the received data for data integrity.
- **Authentication:** Data is received only from the permitted endpoint.
- **Anti-replay:** Even if a packet is encrypted and authenticated, an attacker could try to capture these packets and send them again. By using sequence numbers, IPSec ensures no duplicate packets are transmitted.

Protocols used by IPSec

IPSec is not a protocol; rather, it's built with a suite of protocols. Please note additional headers are added to the packet, which increases its size. In general, the following protocols make the IPSec suite.

Authentication header (AH):

Authentication headers ensure the data packet is from a trusted source (authentication) and the data is not tampered with (data integrity) but do not provide encryption. The data is hashed, and the host that receives the packet can use this hash to ensure that the payload hasn't been modified in transit. The protocol field of the IP header for AH is 51.

Encapsulating security payload (ESP):

This provides data integrity, encryption, and authentication for payload. The encryption takes place based on the operating mode. ESP adds its own header and trailer in each packet. It also produces a unique sequence identifier for each packet. This identifier then allows a device to determine whether a packet has been authorised or not. Packets that are not authorised are discarded and not given to the receiver. Newer versions of ESP also incorporate AH functionality.

The protocol field of the IP header for ESP is 50.

Internet key exchange (IKE):

Security association (SA) refers to the number of protocols used for negotiating the encryption keys and algorithms. SA helps to encrypt raw data and primarily uses the IKE protocol. Internet key exchange (IKE) is used to negotiate cryptographic keys and algorithms that will be used in a session and provides message content protection. The beauty of this protocol, which is being used for the past 20+ years, is that there is a wide range of encryption methods to choose from, like PKI or any other.

IPSec operating modes

There are two modes used by IPSec to treat packet headers.

Tunnel mode: Here, the additional IP header is encapsulated in the original IP packet including the headers. This mode is generally used for connections between gateways that sit outside the private network. A packet is encrypted as it leaves one network and put inside a new packet, whose destination is the gateway for the target network.

Once it arrives at the gateway, it's decrypted and removed from the encapsulating packet, and sent to the target host on the internal network. The header data about the topography of the private networks is thus never exposed while the packet traverses the public internet.

Transport mode: This mode retains the original packet IP header and encrypts the payload alone. It ensures fast and secure end-to-end communications.

IPSec flow

Detailed information on this subject is given at <https://networklessons.com/cisco/ccie-routing-switching/ipsec-internet-protocol-security>.

IPSec is complex and can be implemented in different ways. IKE is used to establish the connection in two phases.